

Automated Risk Assessment for Client Onboarding in Private Banking

Team LadyBugs, Irene Strauss, Beste Güney, Daria Horica, Alina Yaroshchuk

Our project aims to automate the **accept/reject decision** by detecting **data deficiencies** and **cross-source inconsistencies** across a large volume of onboarding files (10,000+).

We developed a **multi-stage validation system** that combines rule-based logic, expert-guided feature engineering, and machine learning—while prioritizing interpretability and ethical decision-making.

Dataset and Challenges

The dataset consists of 10,000 synthetically generated client profiles, designed to closely simulate real-world onboarding scenarios in private banking. The primary challenges include the high administrative burden, data quality issues, and the need for complex feature engineering. Additionally, decisions in this domain must be **explainable and justifiable**, particularly to ensure fairness, regulatory compliance, and trust in the system.

Proposed Solution: A Multi-Stage Validation System

1. Rule-Based Data Validation (Module 1 – Information Checker)

The first module detects clear data deficiencies using strict, rule-based logic:

- **Hard-coded validators** for critical fields such as name, passport number, and email, requiring exact matches across documents.
- **Format and pattern checks** for fields like phone numbers and postal codes.
- **Immediate rejection** for high-risk violations, such as missing passports or mismatched names.

This module alone filters out approximately **20%** of the dataset by identifying clearly non-compliant cases. Its simplicity and transparency make it highly interpretable and easy to audit.

2. Expert-Guided Feature Engineering (Module 2 – Smart Scoring System)

For clients who pass the initial validation, we apply domain-informed feature engineering to assess **reliability** and **economic soundness**. This includes:

- **Wealth modeling**: Aggregating values like savings, investments, and cash holdings into a total assets figure.
- **Wealth consistency scoring**: Evaluating whether reported income sources (salary, inheritance, etc.) plausibly explain declared assets.
- **Currency normalization**: Proposing a system to adjust for cross-country differences using conversion rates and economic scaling (e.g., GDP-weighted or inflation-adjusted metrics).
- **Profitability estimation**: Modeling investment behavior over different time horizons (short-, medium-, long-term), as well as saving patterns.
- **Behavioral signals**: Capturing features from employment history, education, and regional investment biases (e.g., a notable preference for Swiss markets).

This module transforms raw data into features that help differentiate legitimate clients from potentially unreliable ones.

3. AI-Powered Inconsistency Detection (Module 3 – LLM for Subtle Red Flags)

Initial experiments with text embeddings (e.g., Sentence Transformers) did not yield improvements, likely due to the lack of domain-specific tuning. To address this, we propose:

- **Fine-tuning a lightweight language model** (e.g., TinyLLaMA) on client descriptions to detect nuanced inconsistencies—such as illogical fund origin stories or conflicting financial narratives.
- **Confidence scoring**: Flagging suspicious profiles with associated risk scores derived from the model's internal uncertainty.
- **Sentiment analysis**: Considered as an auxiliary signal, though used cautiously, since overly positive tone could reflect personal bias (e.g., the banker being a friend of the client).

This module would ultimately generate interpretable outputs such as textual highlights or natural-language justifications for why a case is considered suspicious.

Prototype Implementation and Results

Our current prototype includes:

- A fully implemented **Module 1**, written in Python, using exact string matching, regular expressions, and date comparison logic to reject clients with clear rule violations (e.g., mismatched passports, invalid emails).
- A partially developed **Module 2**, with core financial features such as total assets, wealth consistency scores, and profitability indicators already in place.
- Integration with **AutoGluon**, an AutoML framework, where we trained over 100 models and achieved up to **75% accuracy** in predicting client acceptance. With improved feature selection and full LLM integration, we expect significantly better performance.

Conclusion and Next Steps

Our approach shows strong potential to streamline onboarding while maintaining fairness and regulatory transparency. The current prototype already offers:

- **High accuracy**, even in its early stage
- **Full interpretability**, with clear rejection reasons (e.g., "Passport number mismatch", "Inconsistent wealth declaration")
- **Ethical compliance**, by ensuring that automated decisions remain auditable and explainable

Next steps:

- Fully integrate the LLM module to detect subtle textual inconsistencies
- Collaborate with domain experts to refine and extend the feature engineering pipeline
- Develop automated, human-readable rejection reports to support transparent decision-making

In summary, our hybrid system—combining rule-based checks, expert-driven modeling, and AI—can significantly reduce inefficiencies in client onboarding while upholding the standards of ethical and compliant decision-making. We believe this approach is well-suited for real-world deployment and ready for further development.